



Super-resolution reconstruction of spinal cord MRI using deep learning: Application to multiple sclerosis atrophy measurement

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ABSTRACT

Accurate quantification of cervical spinal cord (CSC) atrophy is crucial for monitoring disease progression in multiple sclerosis (MS). However, clinical magnetic resonance imaging (MRI) protocols are typically limited by low resolution (LR) and inter-slice gaps, impairing reliable volumetric assessment. To address these constraints, we introduced a novel three-dimensional (3D) deep learning (DL) super-resolution (SR) model designed to reconstruct high-resolution (HR) CSC volumes from LR, sparsely sampled MRI scans. The model was evaluated on both public and clinical datasets using standard quantitative metrics, and its impact on segmentation performance was further assessed. We applied the model to both longitudinal and cross-sectional MS datasets comprising, respectively, five subjects with multiple follow-up scans over 4 to 6 years, and 23 subjects with a single scan. The 3D spinal cord super-resolution (SCSR) model consistently generated reliable reconstructions with enhanced anatomical details. This approach improves the reliability of the segmentation in initially challenging clinical LR scans by generating high-quality CSC SR images, allowing for more precise quantification of CSC volume. Finally, longitudinal regional and global CSC atrophy were estimated and compared with healthy controls (HCs), providing preliminary insights that will benefit from further validation in larger cohorts.

1. Introduction

Multiple sclerosis (MS) is a chronic, inflammatory, neurodegenerative disorder of the central nervous system, defined by demyelination, axonal loss, and neuronal degeneration [1]. Although cerebral white matter lesions are common in MS, spinal cord (SC) involvement is highly prevalent and a major factor in physical disability and long-term functional decline [2]. Quantifying cervical SC (CSC) atrophy has become a sensitive and specific imaging biomarker [3]. Accurate CSC volume quantification is therefore crucial for prognosis, therapy monitoring, and neuroprotective trial endpoints [4].

Given its clinical relevance, magnetic resonance imaging (MRI) is the gold standard for visualizing CSC pathology and assessing atrophy in MS, primarily by measuring cross-sectional area (CSA) or volumetric changes via image segmentation. Studies confirm strong links between SC atrophy and disability progression in MS [5,6]. Early CSC atrophy quantification depended on labor-intensive manual methods [3]. Over two decades, automated segmentation tools like the Spinal Cord Toolbox (SCT) have greatly enhanced measurement consistency [7]. Using

anatomical priors and intensity patterns, these tools enable precise CSC boundary delineation and convert raw images into quantitative biomarkers [8]. However, despite algorithmic advances and applicability to large studies, precise CSC volume quantification is still limited by fundamental constraints of MRI acquisition. Clinical CSC MRI typically uses multi-slice two-dimensional (2D) sequences with relatively thick axial slices and large inter-slice gaps [9,10]. This reduces scan time and motion artifacts but lowers resolution, causes partial volume effects, blurs structures, and compromises segmentation accuracy [11]. Detecting subtle, clinically relevant changes (< 1%/year) becomes especially difficult.

To overcome these acquisition-related limits, computational super-resolution (SR) methods have emerged to reconstruct high-resolution (HR) images from low-resolution (LR) inputs, restoring anatomical details lost during acquisition [12]. In medical imaging, SR has improved brain, cardiac, and fetal MRI resolution, enhancing diagnostic confidence and quantitative accuracy [13,14]. Deep learning (DL), especially convolutional neural networks (CNNs), represents state-of-the-art in

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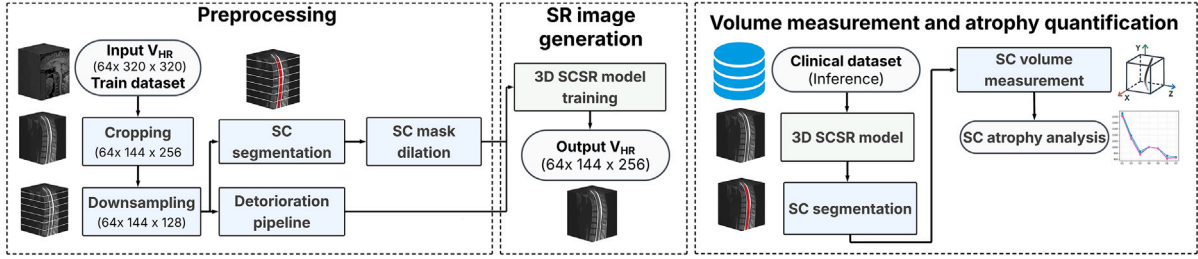


Fig. 1. Overview of the proposed computational framework for CSC analysis.

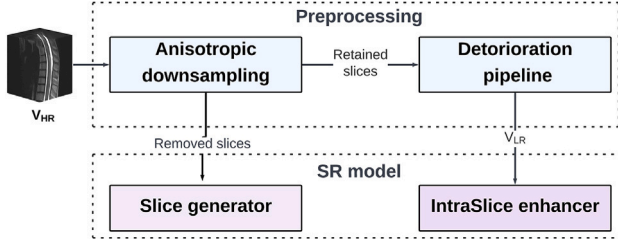


Fig. 2. Illustration of the proposed deterioration and reconstruction pipeline.

SR by learning complex LR-to-HR mappings [15]. While SR has been applied to SC imaging, most work targets specific tasks without fully restoring volumetric integrity from sparse clinical data. For example, Greenspan et al. [16] adapted computer-vision SR using sub-pixel shifts across three 2D image stacks to reconstruct HR volumes, a promising approach later replicated. Other techniques include sparse coding [17], random forests [18], and CNN models [19–21]. Recently, [22,23] have also explored SR's role in improving MS lesion quantification. Despite advances, most MS SC morphometric studies focus on CSA [13,14], with few quantifying SC volume [24,25]. Though CSA is valuable, volume provides a more precise atrophy measure [26]. Valsasina et al. developed an automated pipeline detecting regional atrophy across cervical segments using voxel-based morphometry [27]. Most studies assess SC atrophy at single levels, usually C2 or C3 [25]. No current method reliably reconstructs CSC volume from undersampled clinical images, limiting volume-based atrophy tracking's clinical impact.

To address these challenges, we proposed a 3D SCSR DL framework designed to reconstruct HR CSC volumes from LR clinical MRI. Unlike prior approaches, our model restores complete volumetric detail, accurately reconstructing undersampled axial slices. This enables precise 3D volume estimation and supports longitudinal MS atrophy tracking. By enabling volumetric reconstruction from routine clinical scans, 3D SCSR model bridges research-grade imaging and clinical practice, empowering personalized care and enhancing trial sensitivity to neurodegeneration.

2. Methods

We introduced a novel computational method for 3D reconstruction and volume estimation of CSC MRIs scans as outlined in Fig. 1. We developed a two-phase process: first, training the proposed 3D SR model; then, during the inference phase, applying the trained model to clinical data for accurate volume measurement and CSC atrophy analysis.

2.1. Data description

For training the proposed SR model, we used T1-weighted images (T1-w) from the healthy controls (HCs) subset of the Spine Generic dataset [28], which included 260 HR CSC MRI volumes acquired at 3T

Table 1

Summary of the longitudinal MS dataset.

Subject	Number of scans	Date
Sub-MS-01	5	2021 (Feb), 2021 (July), 2022, 2023, 2024
Sub-MS-02	4	2020, 2021, 2023, 2024
Sub-MS-03	4	2021, 2022, 2023, 2024
Sub-MS-04	7	2016 (May), 2016 (Nov), 2017, 2019, 2020, 2021, 2023
Sub-MS-05	3	2021, 2022, 2023

with 1 mm³ isotropic resolution. Subjects ranged from 19 to 56 years old, including 139 males and 121 females.

We also analyzed local clinical datasets for MS patients. These images were acquired at 1.5T, had voxel sizes primarily around 1.5 mm³ and varied in resolution. All images were anonymized and defaced to maintain confidentiality. The first consists of 5 patients aged 24 to 36 years having multiple scans over time as summarized in Table 1 used for the longitudinal CSC volume study. The second contains 23 additional MS subjects with a single scan used for the cross-sectional study.

2.2. Data preprocessing

A key step in data preparation was the generation of automated CSC binary segmentation masks to guide the model's focus on the CSC region. These masks were dilated to direct the model's attention to both the CSC interior and its boundaries, improving edge representation and enabling their use as an attention masks by weighting the feature maps. Initial binary masks were generated using SCT [7]. The reliability of these automated masks was validated against manually corrected ground truth segmentation masks from the Spine Generic dataset. This validation ensured that the reconstruction process is guided by accurate anatomical boundaries. To further improve the model's robustness under realistic clinical conditions, a comprehensive 3D MRI deterioration pipeline was designed, as detailed in Fig. 2.

The first step simulates anisotropic acquisition by downsampling the HR volume along the through-plane (Z) axis by a factor of 2. Let $V_{HR} \in \mathbb{R}^{H \times W \times D}$ denote the HR volume, where H , W , and D are height, width, and depth (number of axial slices). We generated an LR deteriorated volume $V_{LR} \in \mathbb{R}^{H \times W \times D_{LR}}$, with $D_{LR} = D/2$, by removing alternate axial slices and updating spatial metadata to reflect the new voxel spacing.

The removed slices correspond exactly to those the model's Slice-Generator decoder must reconstruct. The retained V_{LR} slices then undergo further deterioration steps, including Gaussian blur, Gaussian noise, and random affine transformations combined with directional blurring, replicating patient movement during acquisition. The final output of these transformations is an LR image with degraded axial slices. These slices are then enhanced by the model's IntraSliceEnhancer decoder.

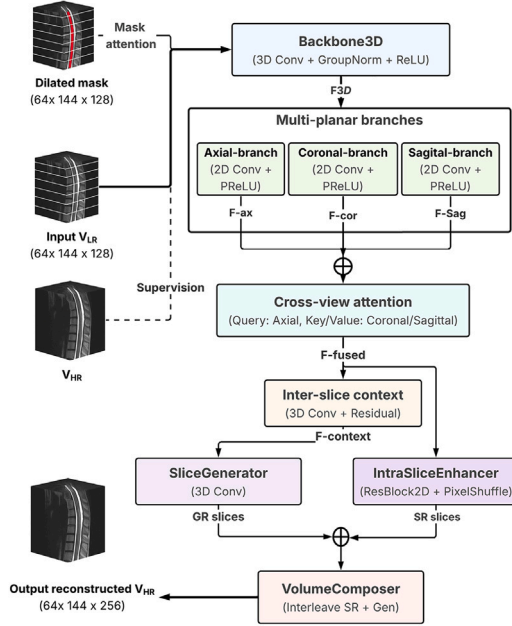


Fig. 3. Schematic representation of 3D SCSR model architecture.

2.3. Super resolution image generation

2.3.1. Model architecture

The proposed 3D SCSR model upsamples input MRI data by a 2×scale along the through-plane axis (z), matching the average 2 mm³ voxel size of our clinical CSC MRI. The model processes the LR volume V_{LR} through a multi-stage architecture to reconstruct the HR volume V_{HR} , as illustrated in Fig. 3.

The proposed model adopts a dual-branch architecture that simultaneously enhances existing slices and generates intermediate slices to improve volumetric resolution. This architecture is complemented by a cross-view attention mechanism that enforces consistency across anatomical planes (axial, coronal, and sagittal). In addition, the framework incorporates a SR strategy designed to reverse realistic MRI acquisition artifacts, along with a mask-guided learning scheme that directs the model to focus specifically on CSC anatomy. Together, these components enable accurate volumetric reconstruction, making routine clinical scans, previously suboptimal for quantitative analysis, suitable for more robust segmentation and volumetric assessment.

First, a Backbone3D encoder extracts volumetric features from V_{LR} using mask-guided attention focused on CSC regions, where the dilated binary mask modulates feature maps via sigmoid-weighted gating. These 3D features are then decomposed into three multi-planar branches using 2D convolutions and PReLU activation function. The axial branch features (F_{ax}) captures intra-slice contrast, while the sagittal and coronal features (F_{sag} , F_{cor}) branches link information across slices. This design decouples texture enhancement from longitudinal consistency, improving both interpretability and anatomical consistency.

Next, a cross-view attention mechanism dynamically fuses multi-planar features by computing attention scores between axial queries and coronal/sagittal key-value pairs. Meanwhile, an inter-slice context module captures longitudinal dependencies via 3D convolutions to maintain coherence across adjacent slices. Subsequently, the model splits into dual decoder pathways: the IntraSliceEnhancer (I-SE) refines retained slices using residual blocks and pixel-shuffle upsampling, while the InterSliceGenerator (I-SG) synthesizes missing intermediate slices from contextual 3D features.

To demonstrate the specific contribution of each architectural component, we conducted an ablation study, evaluating the impact of the

I-SE, I-SG, and the Cross-view Attention mechanism on the overall reconstruction performance.

2.3.2. Model training and optimization strategy

All experiments were conducted on a server (NVIDIA A100 40 GB GPU, dual AMD EPYC CPUs, 256 GB RAM) using Python 3.10, PyTorch 2.1.0, TorchIO (v0.18.83), and SCT (v6.3).

The dataset (N=260) was randomly split into training, validation, and testing sets at 70%, 15%, and 15%, respectively. During training, patches of 4 contiguous axial slices were extracted per sample, with a batch size of 8 (training) and 2 (validation). All MRI volumes underwent per-subject z-score normalization. Input volumes V_{LR} had a simulated resolution 1 × 1 × 2 mm³ and a size of 64 × 144 × 128 voxels (128 axial slices retained after anisotropic downsampling). The SR output V_{SR} was reconstructed at isotropic 1 mm³ resolution, size 64 × 144 × 256 voxels.

Training lasted 120 epochs with early stopping after 15 epochs without validation improvement to prevent overfitting. Extensive data augmentation included random rotations and translations. Slice dropout was implemented to enhance robustness to missing data by randomly dropping entire axial slices during training. peak signal-to-noise ratio (PSNR) [29] was monitored for validation. Hyperparameters were empirically optimized based on validation performance.

To optimize the 3D SCSR model, we defined a hybrid loss function that jointly optimizes both the I-SR and I-GR branches, combining mean absolute error (L_1) and the structural similarity index measure (SSIM) [30].

$$L_{\text{total}} = \lambda_{\text{SR}} \cdot \left(\alpha \cdot L_1^{\text{SR}} + \beta \cdot (1 - \text{SSIM}^{\text{SR}}) \right) + \lambda_{\text{GR}} \cdot \left(\alpha \cdot L_1^{\text{GR}} + \beta \cdot (1 - \text{SSIM}^{\text{GR}}) \right) \quad (1)$$

where $\lambda_{\text{SR}} = 0.6$ and $\lambda_{\text{GR}} = 0.4$ are empirically set weights favoring I-SR during training. This prioritization reflects the direct impact of improved intra-slice resolution on CSC segmentation accuracy and ensures high-quality of slices generated by the I-SG.

With the total loss function L_{total} defined, we implemented a structured optimization approach to facilitate stable training and robust convergence. A two-phase optimization was used. Initially, the Adam optimizer ensured stable early training. Fine-tuning employed AdamW with decoupled weight decay for better regularization, combined with a Cosine annealing learning rate schedule. This included a linear warm-up to a maximum rate of 1×10^{-4} over 30 epochs, followed by cyclic decay between 1×10^{-4} and 1×10^{-6} every 15 epochs ($T_{\text{max}} = 15$), maintaining a minimum rate $\eta_{\text{min}} = 1 \times 10^{-6}$ for continuous gradient updates.

Slice dropout probability followed a triangular schedule: it increased from 0.05 to 0.2 and then decreased to 0.0 for final convergence. This incrementally improved robustness to missing data before focusing on high-fidelity reconstruction. After training, the model was integrated into the clinical workflow for volume quantification and atrophy assessment.

2.4. Volume measurement and atrophy quantification

To accurately quantify CSC volume from 3D MRI scans, we employed a direct volumetric approach using the 3D segmentation mask, calculating CSC volume by multiplying physical voxel volume by the number of voxels in the segmented region. A comparison with SCT results on the Spine Generic dataset showed no significant difference ($p = 0.96$), confirming reliability. This pipeline was then applied to the clinical MS dataset, with quantitative results detailed in the Results section.

Table 2

Quantitative performance of the 3D SCSR model compared with baseline and deep learning methods across different reconstruction tasks on the held-out test set (Mean $\pm$ standard deviation). 'N/A' indicates tasks not covered by the respective methods. Datasets marked with * are open access.

Model	Dataset	Cohort	ROI	I-SE		I-SG		SR-VR	
				PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Trilinear Interpolation [31]	Clinical	MS	SC	N/A	N/A	27.45	0.82	28.10	0.83
B-Spline Interpolation [31]	Clinical	MS	SC	N/A	N/A	28.12	0.84	28.95	0.85
SMORE [32]	Clinical	MS	SC	N/A	N/A	30.92	0.88	30.86	0.89
Feng et al. [33]	fastMRI *	HCS	Brain	31.76	0.70	N/A	N/A	N/A	N/A
Giraldo et al. [34]	ISBI2015 *	MS	Brain/SC	N/A	N/A	N/A	N/A	23.70	0.81
Acharya et al. [35]	Mendeley *	HCS	SC	29.57	0.89	N/A	N/A	N/A	N/A
3D SCSR model (Proposed)	Spine Generic*	HCS	CSC	34.22	0.94	31.50	0.91	33.80	0.93
	Clinical	MS		32.23	0.92	31.12	0.92	31.72	0.92

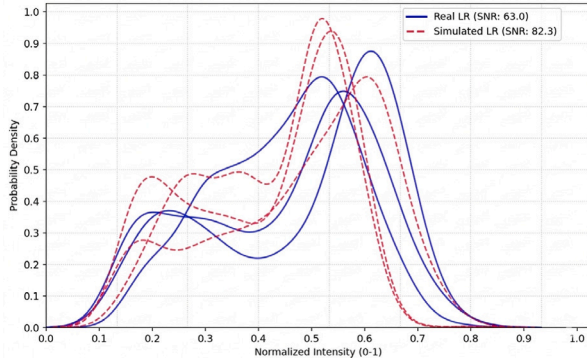


Fig. 4. Comparison of intensity distributions between real and simulated LR MRI at the C3 vertebral level.

3. Results

3.1. Quantitative evaluation

Inference time was measured on a consumer-grade laptop (GTX 1650 Max-Q, 8 GB VRAM). The end-to-end processing of a single clinical CSC volume took 24 ± 4 s, including preprocessing and SR reconstruction.

To evaluate the clinical relevance of the simulated LR image generation process described in Section 2.2, we compared its intensity distributions and resolution characteristics with the real clinical scans. The C3 vertebral level was selected as a common anatomical reference to ensure consistent comparison across subjects. Intensity histograms were computed within a mask-guided region including the CSC and the cerebrospinal fluid (CSF). As shown in Fig. 4, the simulated and real images demonstrate close visual agreement. As well, we compared model performance on simulated LR (SNR ≈ 82) versus real clinical LR (SNR ≈ 63). The Dice score decreased by only 2.1% (0.95 vs 0.93), confirming that the model generalizes well despite SNR differences.

In addition, we evaluated the sensitivity of the model configuration to moderate variations in key hyperparameters, including the loss weights (λ_{SR} , λ_{GR}), the balance between L_1 and SSIM terms, the maximum learning rate, and the slice dropout probability. Each parameter was independently perturbed within ± 10 to 20% of the nominal value while keeping all other settings fixed. Detailed quantitative results are reported in Table S1 (Supplementary Material). Across all tested configurations, performance varied by less than 0.6 dB in PSNR and 0.015 in SSIM, with no degradation in Dice score or visual quality. These results indicate that the proposed model is stable and does not rely on fine-tuned hyperparameter settings.

Table 2 presents the quantitative performance of the proposed 3D SCSR model with comparisons to interpolation baselines and DL methods. Reconstruction metrics for clinical MS cohort were computed across both longitudinal and cross-sectional datasets.

Table 3

Quantitative segmentation results across cervical levels. The table compares performance metrics (Dice score and HD95) for the LR baseline and the proposed SR model. Results are averaged across the clinical dataset (N=28).

Level	Dice		HD95 (mm)	
	LR	SR	LR	SR
C1	0.85	0.94	1.85	1.00
C2	0.84	0.93	1.90	0.98
C3	0.86	0.96	1.70	0.89
C4	0.80	0.95	1.75	0.93
C5	0.83	0.94	2.10	1.02
C6	0.78	0.93	2.35	1.12
C7	0.75	0.92	2.50	1.20
Full Volume	0.82	0.95	2.02	1.02

The 3D SCSR model outperforms interpolation baselines and SMORE across all reconstruction tasks. On the clinical dataset, SR volume reconstruction (SR-VR) yields a gain of 2.8 dB in PSNR compared to B-Spline interpolation. Among individual tasks, I-SE achieves the highest consistency, consistent with its role in enhancing in-plane details. I-SG remains more challenging due to the synthesis of missing slices; performance on clinical data (31.12 PSNR) is slightly lower than on Spine Generic (31.50 PSNR), likely reflecting real-world acquisition variability, yet still surpasses SMORE. When combined (SR-VR), PSNR and SSIM remain high across datasets, indicating robust restoration of anatomical details.

Beyond image fidelity, accurate reconstruction must support reliable anatomical segmentation, a prerequisite for atrophy quantification. To this end, we evaluated segmentation consistency using CSC masks generated by the SCT as automatic references. A single, fixed U-Net was applied identically to LR and SR images, ensuring that observed differences reflect resolution effects rather than segmentation variability.

Segmentation-based metrics confirm improved anatomical consistency (Table 3). Dice scores on SR images exceed 0.92 across all cervical levels (C1–C7), with HD95 values nearly halved compared to LR (1.02 mm vs. 2.02 mm). For context, trilinear interpolation and B-spline interpolation yield Dice scores of 0.83 and 0.85, while SMORE reaches 0.89, highlighting the advantage of the 3D SCSR model (Dice score = 0.95). These gains persist even in challenging lower segments (C6–C7), which are prone to motion and field inhomogeneities. These results are consistent with the overall architecture of 3D SCSR model, which integrates multiple components to enhance structural definition.

3.2. Qualitative evaluation

The SR slices displayed improved clarity and finer anatomical detail compared to LR inputs. Fig. 5 illustrates representative cases in the sagittal view, comparing the original input volume with the HR output from the 3D SCSR model. A zoomed-in view highlights enhanced CSC

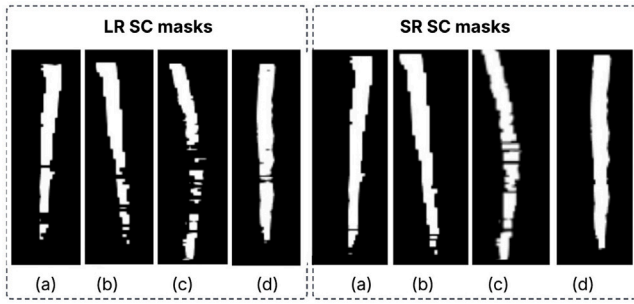


Fig. 5. Visual comparison of sagittal CSC segmentation mask quality before and after application of the SR model, across four distinct subjects.

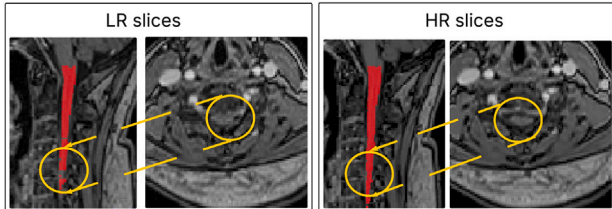


Fig. 6. Visual comparison of CSC segmentation mask quality before and after applying the SR model. The sagittal view shows a discontinuous mask before SR, caused by blurred CSC contours in the axial plane. Post-application, enhanced image quality yields clearer boundaries, resulting in a continuous and anatomically accurate segmentation mask.

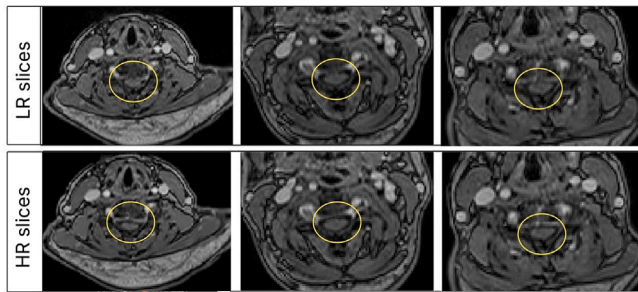


Fig. 7. A comparison of the SC Mask for Subject Sub-MS-01 before and after SR, including a zoomed axial slice that was unsegmented before SR.

contours and surrounding structures, showcasing the model's ability to recover subtle anatomical features lost during the degradation process. To validate these visual improvements, a blind assessment by a radiologist was conducted to compare the quality of SR and LR images. The results show that SR images are consistently superior to LR images across all cases.

A key clinical aim of the proposed model is to enhance CSC segmentation mask quality, which is crucial for atrophy analysis. To validate this, we compared segmentation quality before and after applying the 3D SCSR model. Fig. 6 shows visual examples illustrating the improved mask quality in axial views and Fig. 7 illustrates the practical benefit of SR on segmentation using Subject Sub-MS-01. It compares CSC segmentation masks from the original LR and SR images. The zoomed-in axial slice, previously difficult to segment from LR input, became analyzable after SR. This confirms the model's ability to restore critical anatomical details lost in LR scans.

To assess lesion preservation, we compared lesions between original LR scans and SR images in the MS subjects. As shown in Figs. 8, lesions in axial and sagittal views were enhanced from original LR to SR scans. Results show that the 3D SCSR model generalizes to pathological data and preserves lesions.

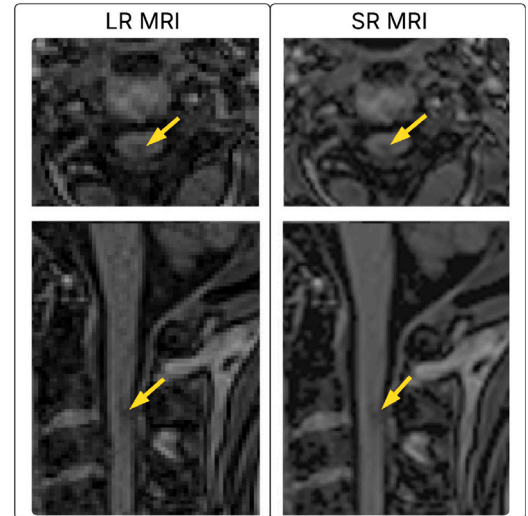


Fig. 8. Visual comparison between lesions in the original LR scan and its corresponding SR scan.



Fig. 9. 3D surface rendering of the CSC segmentation mask.

The 3D reconstruction of the CSC segmentation mask (Fig. 9) further supports volume accuracy.

3.3. Ablation study

Quantitative ablation results (Table 4) confirm that each architectural component plays a distinct role. The I-SE is the most critical for image quality, as its removal caused the most significant drop in PSNR and SSIM. This demonstrates that enhancing local spatial features within each slice is vital for high-quality 3D reconstruction. In contrast, the exclusion of the I-SG primarily affected volumetric continuity by reducing the consistency of intermediate slice generation, confirming its role in maintaining a coherent 3D reconstruction. Regarding anatomical precision, the removal of the mask-guided attention and the integrated multi-planar branches with cross-view attention led to notably decreased Dice scores (0.90 and 0.91, respectively, from 0.95 in the full model). These results highlight the importance of spatial masking for regional focus and multi-view fusion for maintaining structural consistency. Overall, the full model yielded the highest performance, confirming that the synergy between dual-pathway reconstruction and multi-stage attention is key to optimizing CSC restoration and segmentation results.

3.4. Volume measurement and atrophy analysis

Building on the qualitatively confirmed anatomical accuracy and segmentation utility of the 3D SCSR model output, we applied the

Table 4

Ablation study of the proposed model components. Each row represents the full model with the specified component removed.

Model configuration	PSNR (dB)	SSIM	Dice
Full Proposed Model	33.80	0.93	0.95
-SE decoder	30.12	0.85	0.89
I-SG decoder	31.40	0.88	0.91
Mask-guided attention	32.55	0.91	0.90
Multi-planar & Cross-view	31.85	0.88	0.91

pipeline to the MS cohort to quantify total and segmental cervical volumes over multiple time points. Group-wise atrophy results are summarized in Table 5, which reports mean $\pm$ standard deviation for MS patients and HCs, as well as the relative atrophy differences and estimated annual atrophy rates. These results provided preliminary quantitative insights into disease progression trends that will require validation in larger cohorts. Finally, (Table 6) provides the detailed individual cervical level (C1–C7) volumes and total volume for each MS patient across all available time points.

The distribution of CSC volumes across segments followed expected morphological patterns in both cohorts. The upper cervical segment (C1) showed the largest volumes. A decrease was seen at C2 and C3, followed by an increase at the cervical enlargement (C4–C5). In MS patients, the peak volume was at C4 (mean 1026.51 mm³), while HCs peaked at C5 (mean 973.22 mm³), suggesting a slightly different enlargement pattern. Volumes then decreased towards C6 and C7, reflecting anatomical tapering towards the thoracic region.

Cross-sectional analysis using two-sample t-tests compared MS patients and HCs (Table 5), revealing CSC atrophy in MS cohort notably at C2 (1074.32 ± 50.63 vs 1280 ± 102.44 mm³, $p = 0.003$) and C3 (969.45 ± 60.92 vs 1168 ± 70.85 mm³, $p = 0.001$). Total CSC volume was 14.46% lower in MS patients (6803.02 ± 364.27 vs 7953 ± 538.92 mm³, $p < 0.001$). Lower cervical levels (C6–C7) showed no significant atrophy ($p > 0.1$), indicating segment-specific vulnerability with early neurodegeneration mainly at upper cervical regions (C2–C3). These findings suggest that regional atrophy mirrors the global pattern, with C2 and C3 showing the largest absolute volume difference (reduction of 205 mm³ and 198 mm³, respectively). These group comparisons are visually summarized in Fig. 10.

Furthermore, the longitudinal data from individual MS patients (Table 6) suggest a trend towards progressive global atrophy. For subjects with missing segments (marked –), total volume sums available segments, with partial data flagged (* in the Total Volume column). For example, Sub-MS-01 showed a gradual cervical volume decrease from 6978.77 mm³ in 2021 to 6893.69 mm³ in 2024. Similarly, Sub-MS-05's volume declined from 7190.94 mm³ (2016) to 6967.71 mm³ (2023), despite minor fluctuations due to missing data. These annual volume changes confirm global CSC atrophy over time.

In addition, in patients with global volume reduction, regional atrophy progressed over time. For example, Sub-MS-01's C2 volume declined from 1026.76 mm³ (2021) to 999.79 mm³ (2024), highlighting localized volume loss. (Table 6) shows patient-specific atrophy per cervical level. While patterns align with group findings, formal longitudinal modeling is needed for statistical validation.

Overall, our SR pipeline: (1) consistently improves image quality over native LR inputs (PSNR gain > 3 dB), (2) enables previously unreachable segmentation, and (3) provides exploratory insights into global and regional CSC atrophy patterns in MS cohort, with up to 14.5% total cervical volume loss relative to HCs.

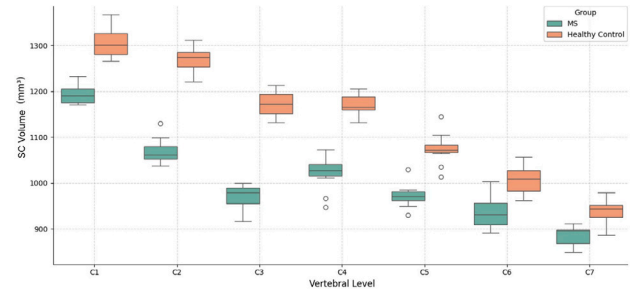


Fig. 10. Box plots comparing CSC volumes between MS patients and HCs per vertebral levels (C1–C7).

4. Discussion

This study introduced the 3D SCSR model, a new approach for CSC MRI SR addressing key needs for improved image quality and quantitative analysis. LR CSC MRI often limits precise diagnosis and longitudinal monitoring in diseases like MS. Our 3D SCSR model converts standard scans into SR images, improving segmentation accuracy and enabling reliable CSC volume measurement and atrophy assessment. Its 3D architecture captures continuous spatial context across entire volumes, surpassing traditional 2D slice-by-slice methods for more coherent reconstructions.

SC imaging research is scarce compared to brain studies, especially in neurodegeneration. Existing work mostly treats SR separately or focus on CSC atrophy quantification from native-resolution MRI. Toufani et al. [24] combined SR and clinical assessment but relied on interpolation, analyzed only C2–C3, lacked longitudinal data, and compared MS with Neuromyelitis Optica without HCs, limiting generalizability. Dewey et al. [13] enhanced 2D SC MRI resolution using zero-shot SR (SMORE), validated for MS atrophy at the upper cervical cord without volumetric analysis.

Unlike prior methods focusing solely on inter-slice generation or intra-slice enhancement [35], our model integrates both with a dual-branch architecture and volume composition, producing a final 3D SR volume ready for clinical use. Compared to mainly intra-slice 2D methods, 3D SCSR model achieves higher PSNR (e.g., 34.62 dB in Ayaz et al. [15]). Though dataset differences limit direct comparisons, robust performance on both the Spine Generic and the clinical dataset underscores clinical utility.

While PSNR and SSIM could improve, the model prioritizes CSC and boundary precision over surrounding tissue, fitting its clinical goal of accurate segmentation and volume quantification. By enabling accurate segmentation from routinely acquired low-resolution scans, 3D SCSR model unlocks quantitative atrophy measurement in standard clinical workflows, without requiring any protocol changes. In settings where accelerated acquisitions are available, the framework is also compatible with faster LR acquisition sequences; as prior work has shown, SR can reduce MRI scan times by 20%–40% without compromising diagnostic utility [36]. Integrating such quantitative analysis into clinical protocols could enhance patient care.

CSC volumes across levels were evaluated on the full clinical dataset, including 5 subjects with longitudinal follow-up and 23 additional subjects with a single scan, providing a broader group-level assessment. In contrast, annual atrophy rates were computed only for the available subset of 5 subjects with repeated scans over time.

Clinically, 3D SCSR model provides a preliminary framework for quantifying neurodegeneration markers: a 14% CSC volume reduction in MS patients (7078 mm³) versus HCs (7953 mm³) highlights the potential of SR-enhanced imaging to capture trends of CSC atrophy, even in LR clinical datasets. Cohort differences explain some volume discrepancies with other studies like [25]. Our C2–C3 MS volumes (2043 mm³) align well with [24,27], affirming regional measurement

Table 5

Group-wise average CSC volumes (mm^3) per segment (C1–C7) and total cervical volume, including mean $\pm$ standard deviation for MS patients (N=28) and HCs (Spine Generic) as well as the 95% confidence intervals (CI) for each group. Relative differences (%) and estimated annual atrophy rates (%/year) are also reported.

	Volume (mm^3)							CSC
	C1	C2	C3	C4	C5	C6	C7	
MS Patients	1190.30 ± 43.67	1081.01 ± 48.16	968.43 ± 59.86	1028.11 ± 68.84	975.91 ± 32.77	937.91 ± 26.08	896.53 ± 25.89	7078.20 ± 115.31
MS 95% CI	[1131.98, 1238.44]	[1011.45, 1137.19]	[893.81, 1045.09]	[943.55, 1109.47]	[935.14, 1011.30]	[877.64, 997.48]	[832.91, 944.45]	[6350.72, 7255.32]
HCs	1324.00 ± 142.23	1280.00 ± 102.44	1168.00 ± 70.85	1177.00 ± 87.54	1071.00 ± 56.09	1011.00 ± 53.76	922.00 ± 43.98	7953.00 ± 538.92
HC 95% CI	[1263.94, 1384.06]	[1236.74, 1323.26]	[1138.08, 1197.92]	[1047.32, 1094.68]	[988.30, 1033.70]	[903.43, 940.57]	[7725.43, 8180.57]	
Difference (%)	−10.10	−15.55	−17.09	−12.65	−8.88	−7.23	−2.76	−11.00
Atrophy (%)	0.52	0.70	1.13	0.38	0.24	0.41	0.14	0.41

Table 6

Longitudinal CSC volume measurements (mm^3) per vertebral level (C1–C7) and total cervical volume in individual MS patients (N=5) from the Private Dataset. Measurements are reported over multiple time points. Missing segment data are indicated by “–”.

Subject	Date	Volume (mm^3)							Total CSC
		C1	C2	C3	C4	C5	C6	C7	
Sub-MS-01	02–2021	1143.83	1034.47	1045.64	959.67	981.89	937.60	875.67	6978.77
Sub-MS-01	07–2021	1142.76	1026.76	1035.95	959.49	981.70	937.24	–	–
Sub-MS-01	2022	1137.01	1023.11	1033.22	957.34	980.68	937.11	–	–
Sub-MS-01	2023	1118.84	1020.48	1033.08	956.77	980.55	936.75	875.61	6921.08
Sub-MS-01	2024	1111.98	999.79	1033.02	956.76	980.40	936.75	874.99	6893.69
Sub-MS-02	2020	1207.98	1148.28	1026.32	1141.90	1010.68	977.16	864.51	7376.83
Sub-MS-02	2021	1200.88	1110.73	1015.28	1133.33	1005.24	968.42	–	–
Sub-MS-02	2023	1200.76	1102.28	1005.72	1132.10	1003.33	967.87	859.65	7271.71
Sub-MS-02	2024	1190.24	1082.14	991.71	1120.97	1002.82	965.34	859.64	7212.86
Sub-MS-03	2021	1178.78	1097.10	1065.74	1001.26	989.89	929.26	919.49	7181.52
Sub-MS-03	2022	1163.25	1084.79	1051.09	1000.80	989.09	912.87	914.87	7116.76
Sub-MS-03	2023	1154.45	1078.73	1044.41	1000.66	988.51	911.19	911.35	7089.30
Sub-MS-03	2024	1152.87	1067.18	1037.96	1000.13	987.26	905.27	911.14	7061.81
Sub-MS-04	06–2016	1256.63	1163.78	921.11	1030.13	996.16	–	–	–
Sub-MS-04	11–2016	1251.12	1160.42	915.50	1028.93	994.79	949.32	890.86	7190.94
Sub-MS-04	2017	1236.81	1149.19	914.09	1018.12	981.02	948.44	890.28	7137.95
Sub-MS-04	2019	1213.12	1128.42	908.50	1012.93	974.79	949.32	890.20	7077.28
Sub-MS-04	2021	1211.35	1113.95	901.38	1005.62	962.00	–	–	–
Sub-MS-04	2022	1207.07	1109.99	894.54	1000.24	960.93	–	–	–
Sub-MS-04	2023	1200.82	1108.77	889.05	994.88	957.68	936.42	880.09	6967.71
Sub-MS-05	2021	1145.92	1037.34	925.15	994.59	913.80	–	–	–
Sub-MS-05	2022	1136.35	1032.32	912.44	990.06	902.53	–	–	–
Sub-MS-05	2023	1121.41	1011.39	903.43	986.22	900.61	876.39	–	–

validity. The annual C2 atrophy rate of 0.72% matches [8], validating clinical sensitivity.

Despite these encouraging results, several methodological limitations must be acknowledged. Limitations included the small number of subjects with longitudinal follow-up (5 patients), which restricts the generalizability of the estimated annual atrophy rates, as well as missing data at C6–C7 for some subjects, which may lead to an underestimation of total CSC atrophy. Future work with data imputation and larger cohorts are needed. Although SR improved segmentation, some slices still failed, motivating plans to expand training data and develop segmentation models tailored for low-quality clinical MRIs. In addition, further research exploring 3D SCSR model effects on other MS biomarkers like lesion volumes will improve the work. Exploring and integrating the pipeline into clinical workflows could enhance the accuracy and efficiency of atrophy monitoring. While this study specifically addresses T1-w images for atrophy measurement, the 3D SCSR model is architecturally sequence-agnostic, and its performance on other contrasts such as T2-weighted or FLAIR MRI will be the subject of future validation.

Beyond MS, the generalizability of the framework deserves consideration. While this study focused on MS, the 3D SCSR model is

architecturally designed to restore generic anatomical structures rather than pathology-specific features. However, the performance of the model in cases of severe CSC deformation, such as in traumatic injury or extrinsic compression, remains to be evaluated. Although the technical precision of the tool is established, further prospective studies are needed to evaluate its integration into routine clinical practice and its role in guiding therapeutic choices for MS patients.

In addition, while the clinical datasets lack diversity in scanner vendors, the 3D SCSR model was trained on the multi-vendor Spine Generic dataset (Siemens, GE, and Philips) to ensure compatibility. Future studies should include a broader range of hardware to further confirm robustness against inter-vendor variability.

5. Conclusion

This work aimed to develop an advanced DL model for the reconstruction of CSC MRI, addressing the limitations of clinical LR scans. Our method introduces a novel 3D framework that synthesizes HR volumetric data from undersampled LR images, enhancing anatomical details. The primary objective is to enable accurate segmentation of the

CSC and a precise quantification of its volume. Comparison with the HCs cohort reveals longitudinal CSC atrophy in cervical enlargement for MS patients. This study improves the quality of MRI by bridging the gap between research-grade imaging and practical clinical applications. Future work will focus on expanding the diversity of the dataset, integrating the model into clinical workflows, and exploring its utility to monitor additional neurodegenerative biomarkers to improve diagnosis and personalized patient care.

CRedit authorship contribution statement

Tayssir Boushila: Methodology, Formal analysis, Data curation, Conceptualization. **Mouna Sahnoun:** Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Fathi Kallel:** Validation, Supervision, Methodology, Data curation, Conceptualization. **Salma Sakka:** Validation, Resources, Data curation. **Mariam Damak:** Validation, Resources, Data curation.

Model availability

The 3D SCSR model is available from the corresponding author upon reasonable request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.bspc.2026.110588>.

Data availability

The authors do not have permission to share data.

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